Introduction

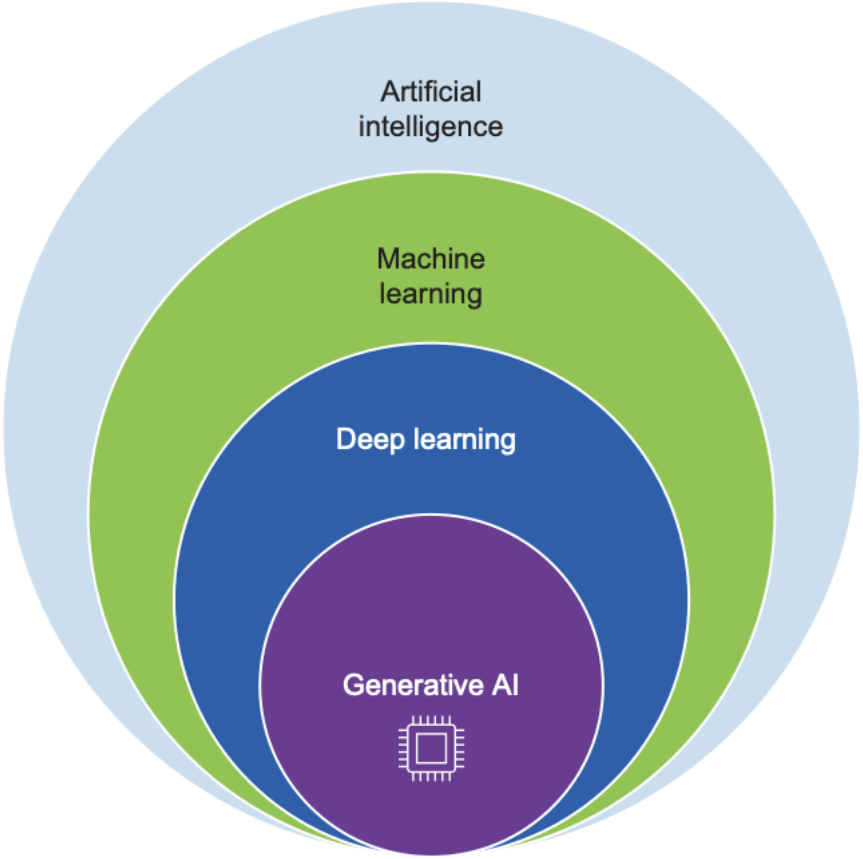
Introduction

- Artificial Intelligence
- Recent History of AI
- Geneartive AI
- LLM Foundation Model
- Popular Gen AI Models
- LLM Configuration Settings

Artificial Intelligence

- “Artificial intelligence is the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence, but AI does not have to confine itself to methods that are biologically observable.”

Generative AI



Generative AI (Large Language Models)

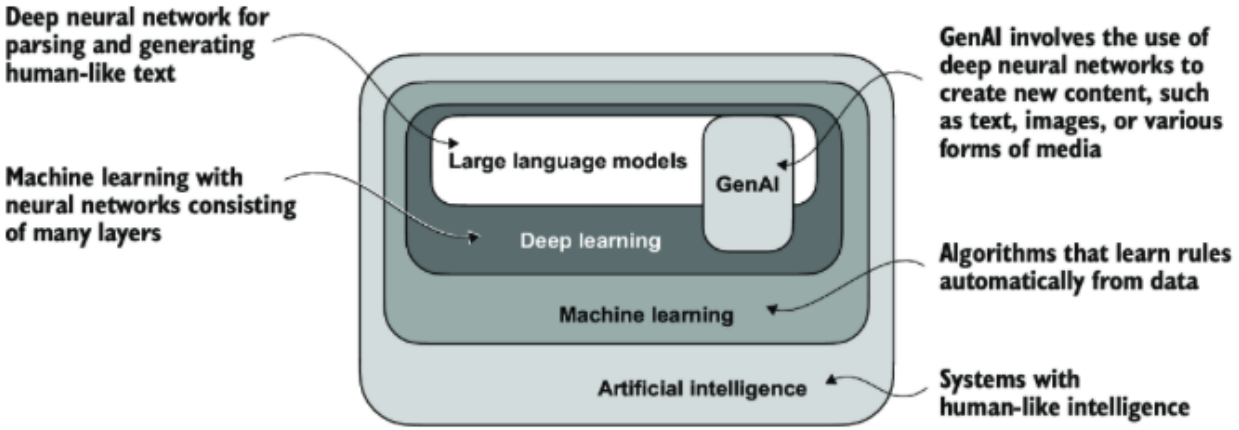


Figure 1.1 As this hierarchical depiction of the relationship between the different fields suggests, LLMs represent a specific application of deep learning techniques, using their ability to process and generate human-like text. Deep learning is a specialized branch of machine learning that focuses on using multilayer neural networks. Machine learning and deep learning are fields aimed at implementing algorithms that enable computers to learn from data and perform tasks that typically require human intelligence.

Recent History Of AI

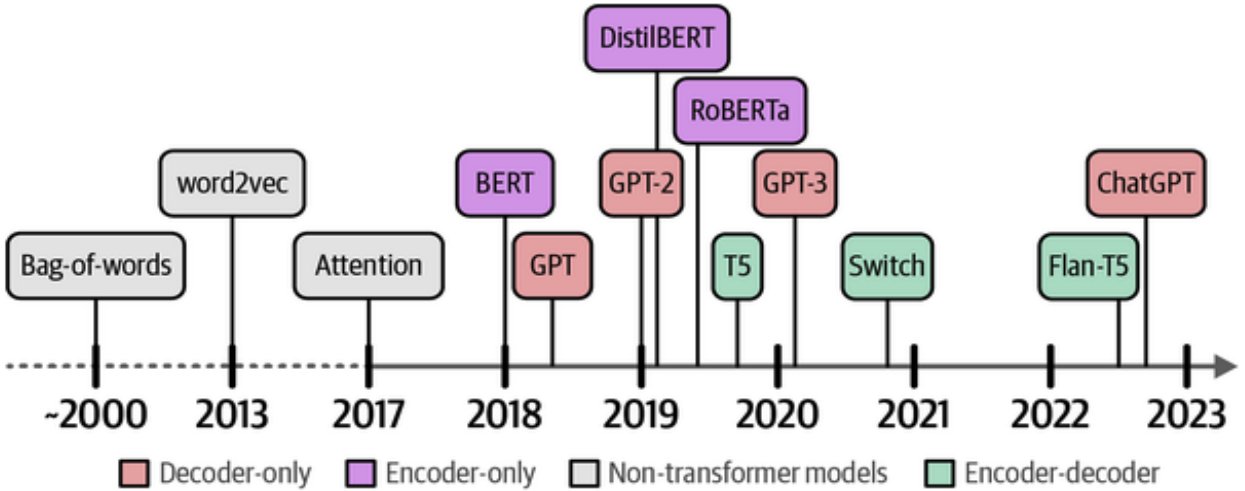
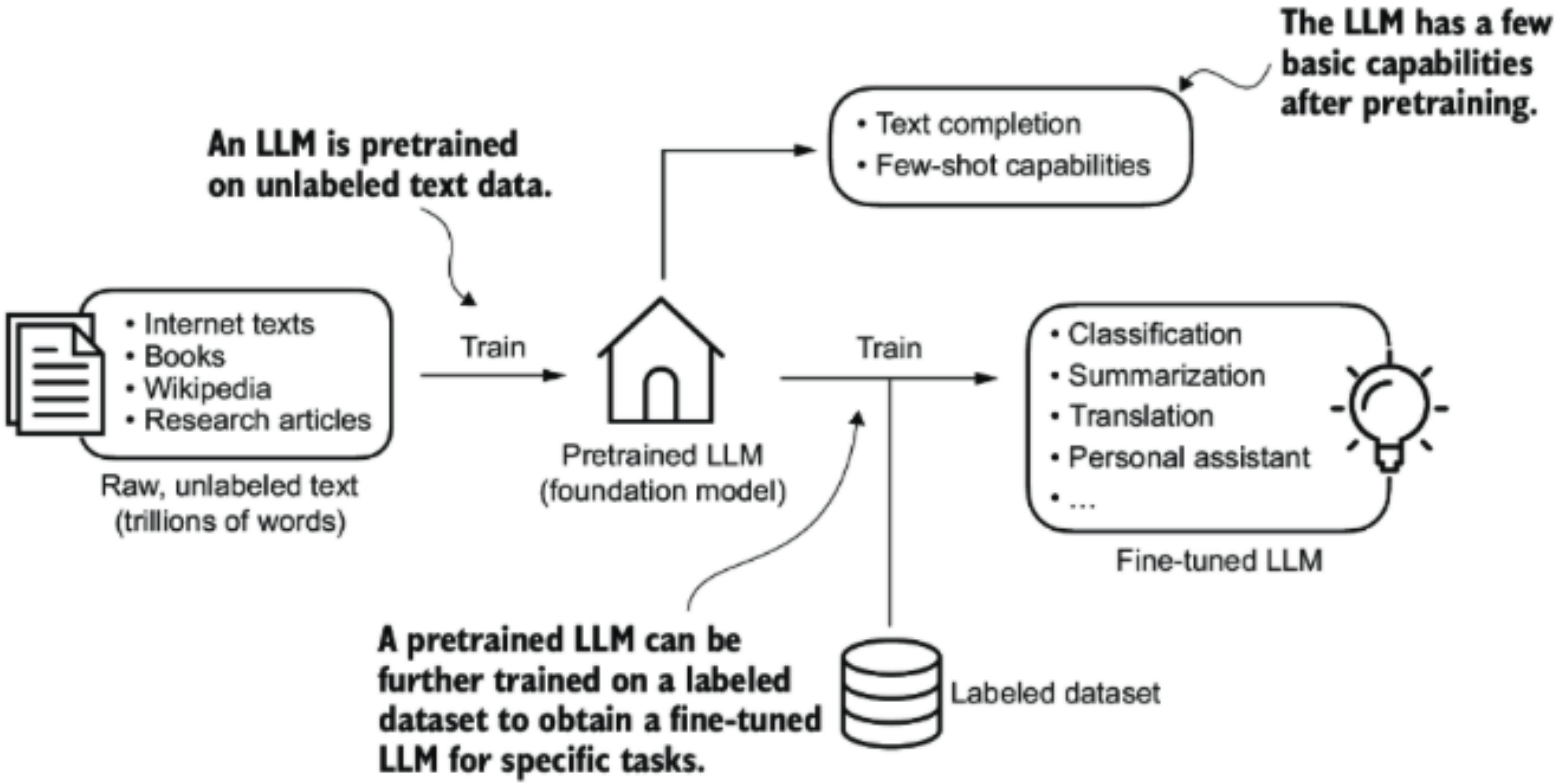


Figure 1-1. A peek into the history of Language AI.

LLM (Foundation Model)



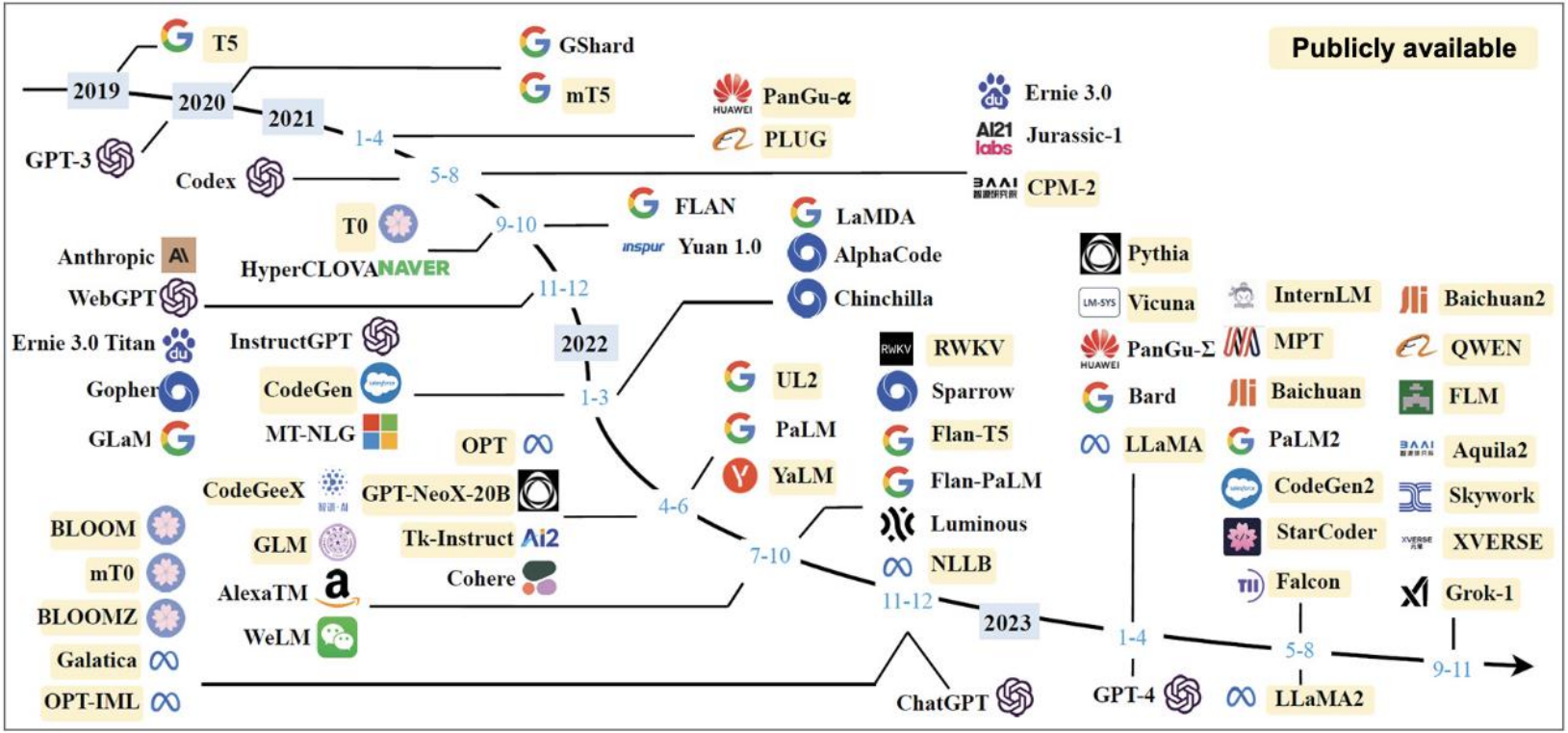
Popular GenAI Models

Name	Description	Area
Generative Pre-trained Transformer (GPT)	A large language model developed by OpenAI and trained on a massive dataset of text and code can generate text, translate languages, write various kinds of creative content, and answer your questions informatively. GPT4-Omni (more commonly referred to as GPT-4o) is a multimodal model. At the time of writing, it is the latest version and is a significant upgrade from GPT-4, offering speed, cost, and capability improvements.	Language/ multimodal
Llama 3	Meta recently released the third version of a natural large language model, open-sourced under a special license. The models come in various sizes and have varying capabilities.	Language
Claude 3	Anthropic has introduced the Claude 3 model family, which includes Claude 3 Haiku, Claude 3 Sonnet, and Claude 3 Opus. These models offer a range of capabilities, with Opus being the most intelligent. It is capable of complex tasks and exhibits near-human comprehension and fluency levels. Like OpenAI's ChatGPT, Claude can generate text, write code, summarize, and reason, among other things, for a given prompt.	Language
Cohere Command	Cohere offers two models (Command R and Command R+) as part of its Command family. While these LLMMS are optimized for various use cases, Cohere's newest large language model, Command R+, is optimized for conversational interaction and long-context tasks. It is designed to be highly performant for complex retrieval-augmented generation (RAG) workflows and multistep tool use.	Language

Popular GenAI Models

Name	Description	Area
Mistral	Mistral Large Language Models are advanced AI models designed for text generation and other language tasks. They have models in different sizes from a collection of open source models (Mistral-7B, 8x7B, and 8x22B) and optimized commercial models (Mistral Small, Medium, and Large), each tailored for different reasoning complexities and workloads.	Language
Gemini	Gemini is Google's new multimodal model that can understand text, images, videos, and audio. It will be available in different sizes (Ultra, Pro, and Nano), each with different capabilities.	Language/ multimodal
DALL-E	Visual AI model developed by OpenAI that can create realistic images from text prompts	Image
Stable Diffusion	Open source image generation model that generates images from a prompt as input. It is primarily used to generate detailed images conditioned on text descriptions and can also be applied to other tasks such as inpainting, outpainting, and generating image-to-image translations.	Image
Midjourney	An image generation model using natural language prompts from a startup called Midjourney, Inc., similar to OpenAI's DALL-E and Stable Diffusion.	Image
CodeWhisperer	CodeWhisperer is an AWS code-generation model that can generate code in several programming languages, including Python, Java, JavaScript, and TypeScript.	Code
CodeLlama	CodeLlama is a large language model built on Llama 2 and specifically trained on code. It is available in various sizes and supports multiple popular programming languages.	Code
Codex	A large language model is trained specifically on code and used to help with code generation. It supports over a dozen programming languages, including some of the more commonly used, such as C#, Java, Python, JavaScript, SQL, Go, PHP, and Shell, among others.	Code

Open Source vs Commerical LLMs



LLM Interface

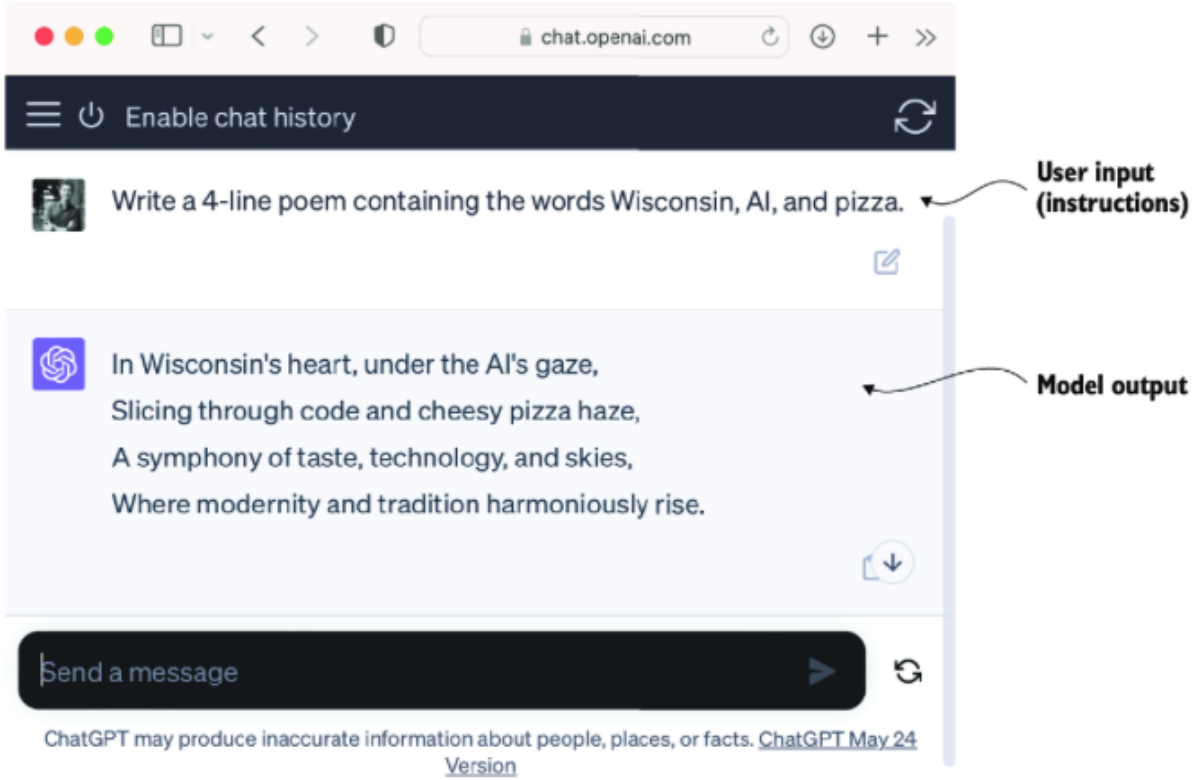
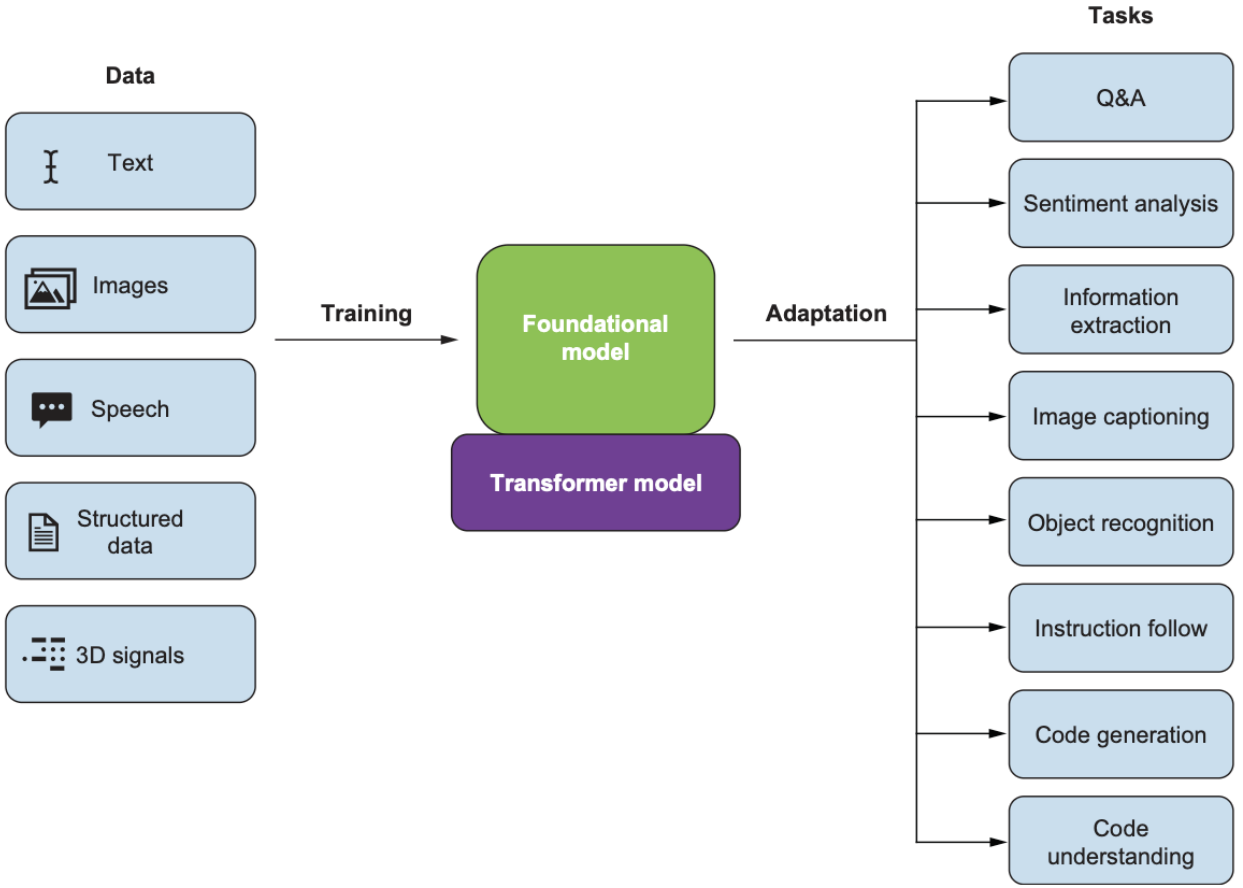


Figure 1.2 LLM interfaces enable natural language communication between users and AI

Foundation Model



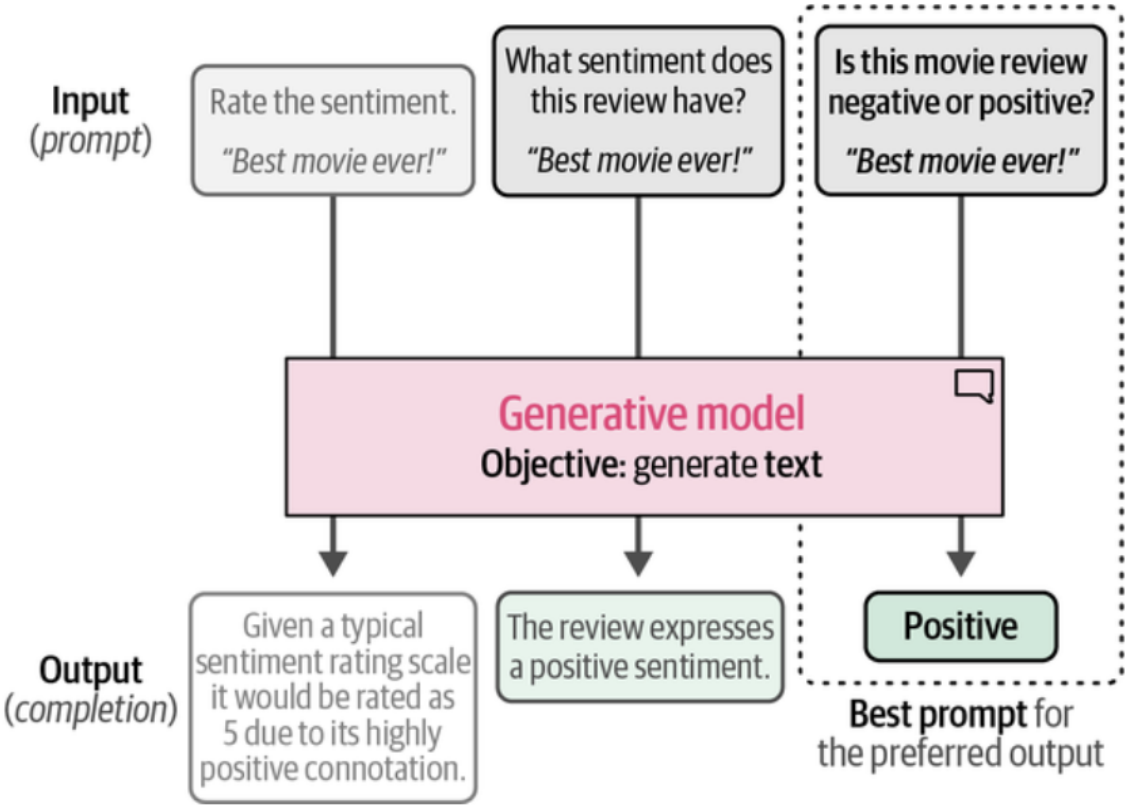


Figure 4-18. Prompt engineering allows prompts to be updated to improve the output generated by the model.

Foundation Model



“decoder-style models like GPT generate text by predicting text one word at a time, they are considered a type of **autoregressive model**. Autoregressive models incorporate their previous outputs as inputs for future predictions. Consequently, in GPT, each new word is chosen based on the sequence that precedes it, which improves the coherence of the resulting text”

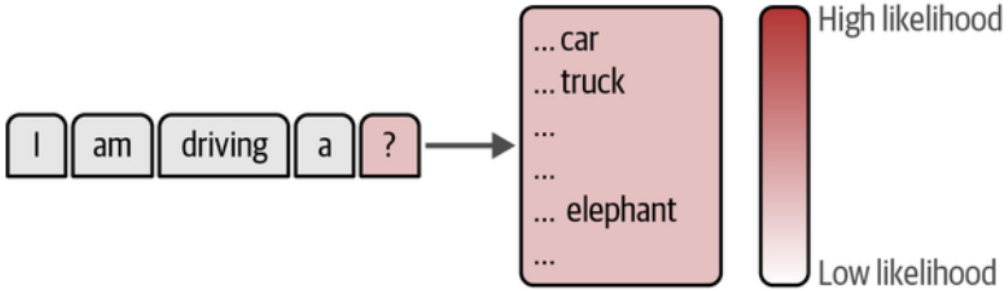


Figure 6-3. The model chooses the next token to generate based on their likelihood scores.

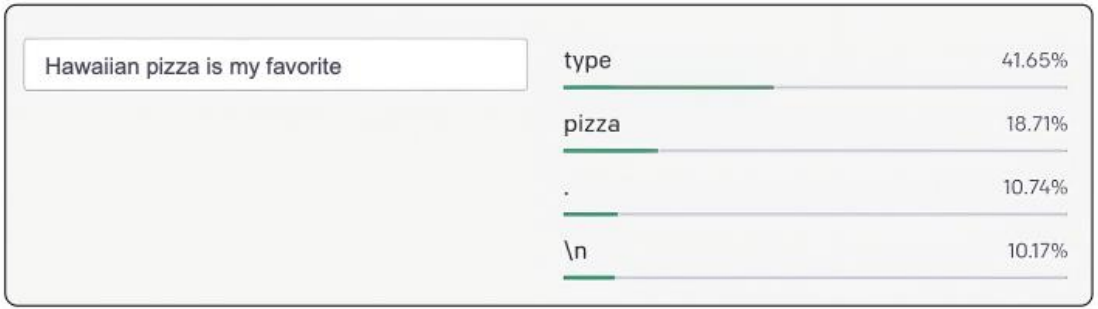
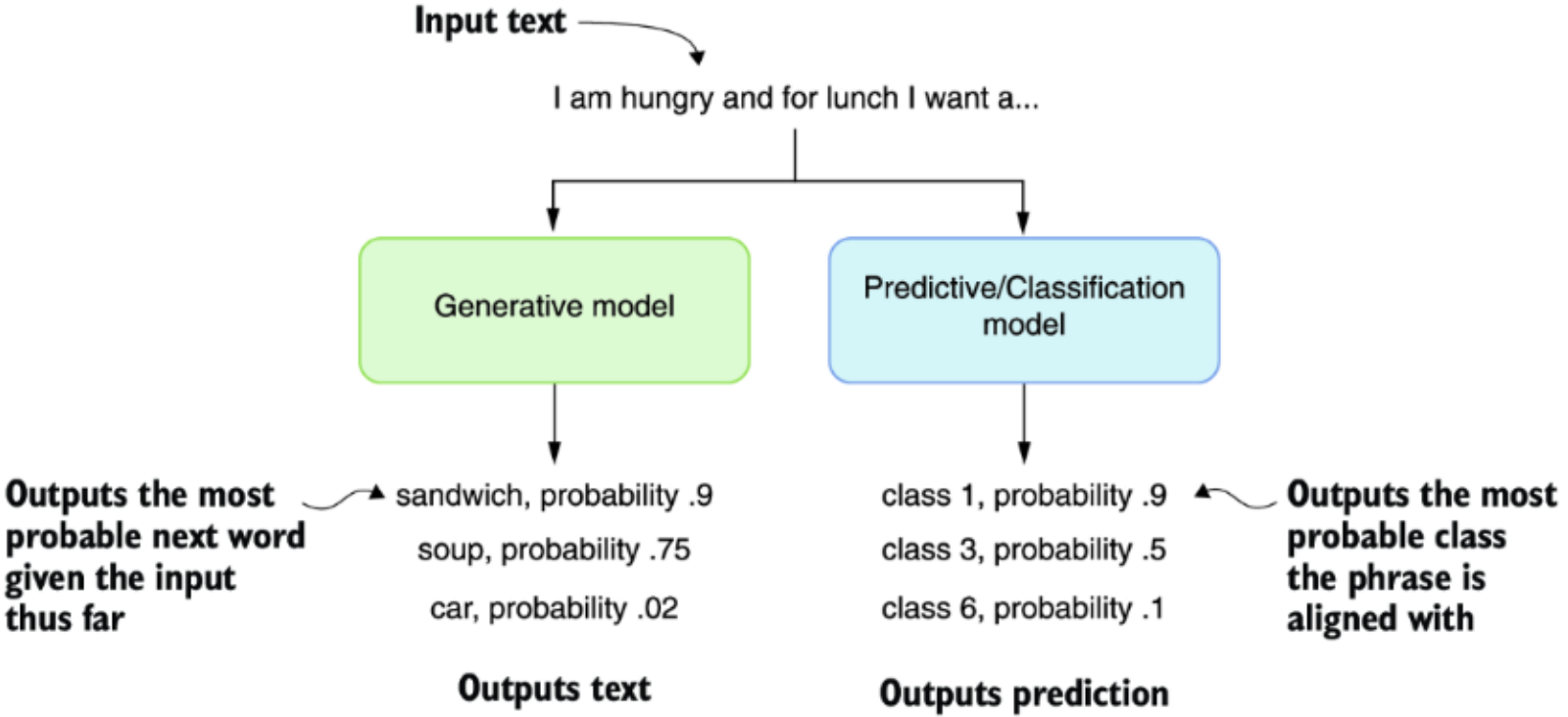


Figure 2.6 Next token probability distribution


Input text
I am hungry and for lunch I want a...
Generative model
Predictive/Classification model
Outputs the most probable next word given the input thus far
sandwich, probability .9
soup, probability .75
car, probability .02
Outputs text
class 1, probability .9
class 3, probability .5
class 6, probability .1
Outputs prediction
Outputs the most probable class the phrase is aligned with

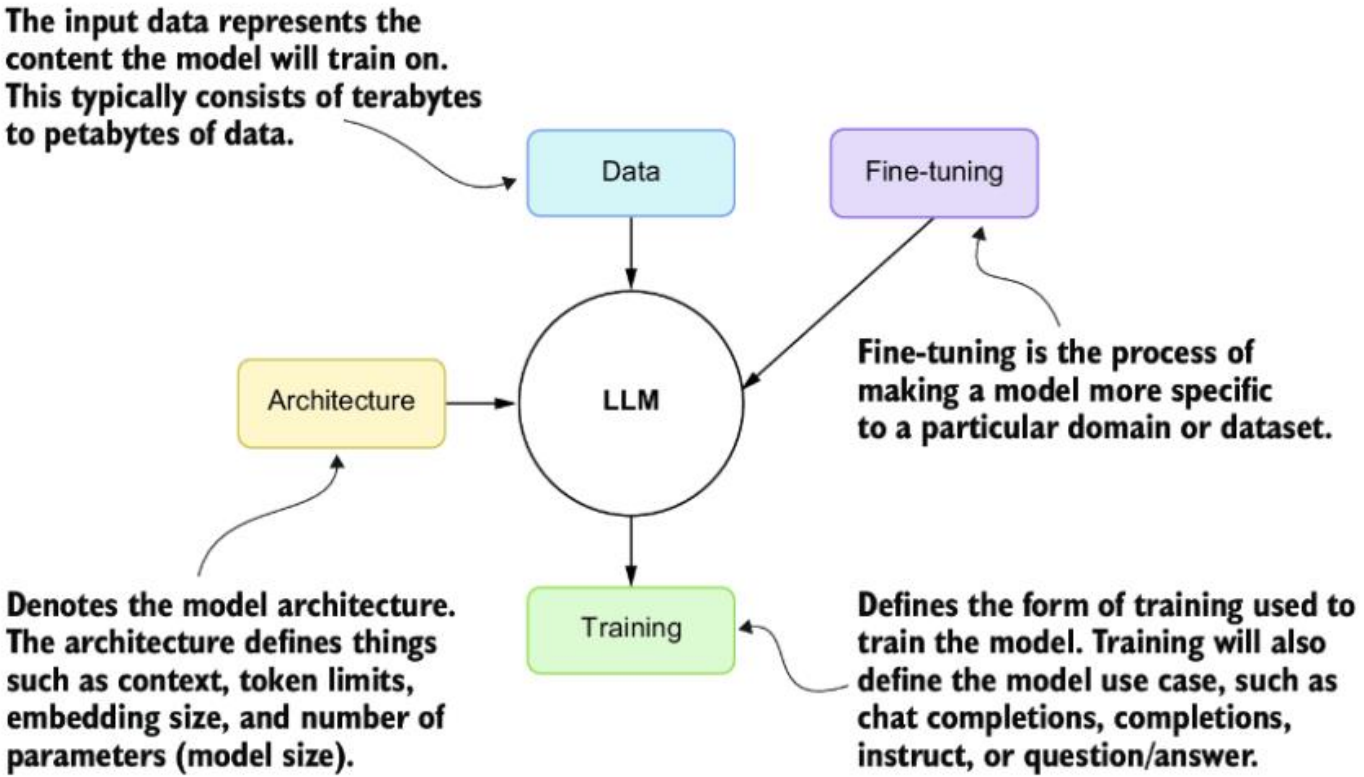

The input data represents the content the model will train on. This typically consists of terabytes to petabytes of data.
Data
Fine-tuning
Architecture
LLM
Fine-tuning is the process of making a model more specific to a particular domain or dataset.
Denotes the model architecture. The architecture defines things such as context, token limits, embedding size, and number of parameters (model size).
Training
Defines the form of training used to train the model. Training will also define the model use case, such as chat completions, completions, instruct, or question/answer.

Table 2.4 LLM configuration settings

Configuration	Description
Max tokens	This sets a limit on the number of tokens per model response. Depending on the model, a maximum limit is shared between the prompt (including system message, examples, message history, and user query) and the model's response.
Temperature	This controls randomness. Lowering the temperature means the model produces more repetitive and deterministic responses. Increasing the temperature results in more unexpected or creative responses. Try adjusting the temperature or `top_p` but not both. Typically, as sequences get longer, the model naturally becomes more confident in its predictions, and one can use a much higher temperature for long prompts without going off-topic. Conversely, using a high-temperature setting on short prompts can lead to unstable outputs.
Top probability (`top_p`)	This is a probability threshold that, like temperature, controls randomness but uses a different method. Lowering `top_p` narrows the model's token selection to the most likely tokens and ignores the long tail of less likely tokens. Increasing `top_p` will allow the model to choose from high- and low-likelihood tokens. Try adjusting either temperature or `top_p`, but not both.
Stop sequences	This makes the model end its response at a desired point. The model response will end before the specified sequence so that it won't contain the stop-sequence text.
Frequency penalty	This reduces the chance of repeating a token proportionally based on how often it has appeared in the text. This decreases the likelihood of repeating the same text in response.
Presence penalty	This reduces the chance of repeating any token that has appeared in the text so far. This increases the likelihood of introducing new topics in a response.

Table 11.3 LLM inference metrics

Metric	Definition
Time to first token (TTFT)	Measures the time it takes for the model to generate the first token after a user query. Lower TTFT means a more responsive user experience. TTFT is influenced by the time required to process the prompt and generate the first output token.
Time-per-output token (TPOT)	Calculates the time required for the model to generate one token for a specific query. Lower TPOT means faster text generation. The model size, the hardware configuration, and the decoding algorithm influence TPOT.
Latency	This metric measures the time it takes for data to move from its starting point to its destination. In the case of LLMs, it is the time for the model to generate a response to the user. The model and the tokens generated influence LLMs' latency. Generally, most of the time is spent generating complete tokens, which are generated one at a time. The longer the generation, the higher the latency.
Throughput	Measures the amount of data that can be transferred in a unit of time. In this case, the number of output tokens per second on a deployment unit can be served across all requests.
Request per second (RPS)	RPS measures the throughput of LLMs in production and indicates the number of requests an LLM can handle every second. This metric is crucial for understanding the scalability and efficiency of LLMs when deployed in real-world applications.

Table 2.1 Types of LLMs

LLM Type	Description
Base LLM	These are the original models, pretrained on a massive corpus of text data, and they can generate text based on the patterns they learned during this training. Some also call these raw language models or even refer to them as foundational models; they can be used out of the box to generate text. They learn powerful and general representations but lack specific expertise in a particular task. GPT-3's DaVinci model is an example of a base LLM.
Instruction-based LLM	This involves using a base LLM and providing explicit instructions in prompt input. In many examples we saw in the previous chapter, we instructed the model to follow instructions, such as "Translate the following text to French:" or "Summarize this article:" Sometimes, these models are also called instruction-tuned LLMs.
Fine-tuned LLM	Fine-tuning involves taking the base LLM and training it further on a task it might perform poorly at, often in a specific domain. An example would be training the model on medical literature if we want it to understand medical topics or training it on customer service interactions if we want it to respond to customer inquiries for a specific industry. Fine-tuning can help make the model more accurate or helpful to particular tasks or domains, but it requires additional data and training time.